

Precise Depth Estimation by Calculating Affine Transformation Parameters

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Abstract— When estimating depth from unmanned aerial vehicle (UAV) consecutive frames, usually only narrow baseline stereo is satisfied. So, the estimation needs a higher image matching accuracy often not achieved in real applications. To tackle this problem, we propose a precise and robust depth estimation scheme: first, by multiview geometry theory, we prove that two corresponding local patches in two consecutive frames satisfy an affine transformation no matter how high the UAV flies and whether it is parallel to the ground or not; second, normal curvatures, local Fourier moment constraints, and spatial smoothness constraints are combined to calculate the affine transformation matrix, and the stepwise least-squares fitting-based PC (SLSF-PC) method is used to calculate translation vectors because it can suppress the illumination effect; third, the normal and the position of the 3-D patch corresponding to two image patches are calculated by the obtained affine transformation matrix and translation vector. The method is very precise and robust due to the combination of 2-D image features and 3-D differential geometry characters such as curvatures and normals. The developed method is tested using real UAV images and images of scale model scenes, confirming the superior performance of the proposed method against the state-of-the-art.

Index Terms— Depth estimation, narrow baseline, stereo vision, unmanned aerial vehicle (UAV) images.

I. INTRODUCTION

STEREO analysis is critical for computer vision applications [1]. Stereo vision is the process of computing the depth of an object in a scene from two or more images taken from different positions [2]. By determining the depth information corresponding to each pixel, it is possible to obtain the 3-D information of a scene, which is useful in many applications [3].

In the case of an ideal stereo geometry, the geometric setting can be represented by a stereo vision model, as shown in Fig. 1, where B is the baseline (distance between two cameras or viewpoints), H is the height of the camera relative to the datum, and h the height of the 3-D point in the scene. The relative distance between the image locations of these

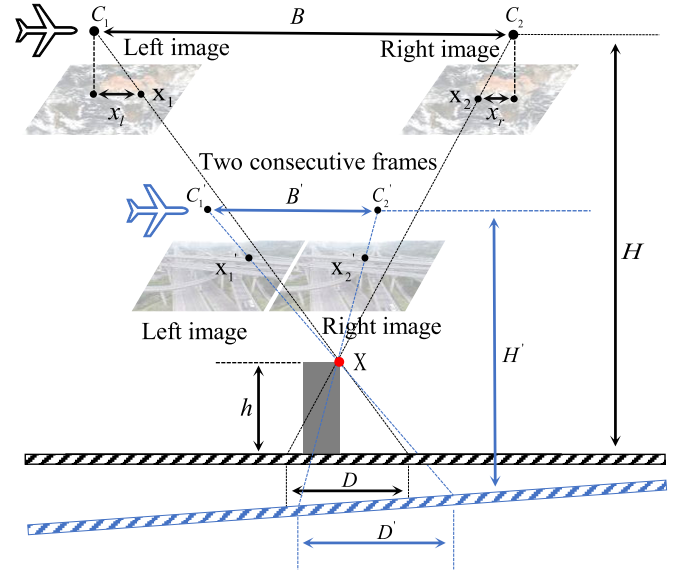


Fig. 1. Diagram showing the process of depth capturing performed by a high-altitude camera and a low-altitude camera. C_1 and C_2 are the centers of the high-altitude camera for two consecutive frames, C'_1 and C'_2 are the centers of the low-altitude camera for two consecutive frames. The 3-D point X is projected onto the 2-D locations x_1 and x_2 in the left and right images taken by the high-altitude camera, and the two 2-D locations x'_1 and x'_2 in the left and right images taken by the low-altitude camera. When the UAV flies high enough, method [4] considers that two corresponding local patches in two consecutive frames satisfy a translation transformation, which requires that all the surfaces in the world be fronto-parallel, i.e., parallel to the image plane. However, even in high-altitude conditions, the camera is not absolutely parallel to the terrain. Therefore, we propose a depth estimation scheme to solve the correspondence problem in the presence of slanted surfaces.

two corresponding points, x_1 and x_2 , is the image disparity ($d = x_1 - x_2$) in pixels. Based on the similar triangles present in this geometry, the disparity D (in ground units) can be calculated [4] as

$$D = \frac{B}{(H - h)}h \quad (1)$$

where $D = dL$, and L is the system resolution or pixel size. Equation (1) can therefore be rearranged as

$$d = \frac{Bh}{(H - h)L}. \quad (2)$$

According to the relationship in (2), the measurement precision of disparity d is proportional to the measurement precision of the baseline B , and inversely proportional to the resolution L . Assuming that the resolution L is limited by hardware constraints, precision of the disparity measurement

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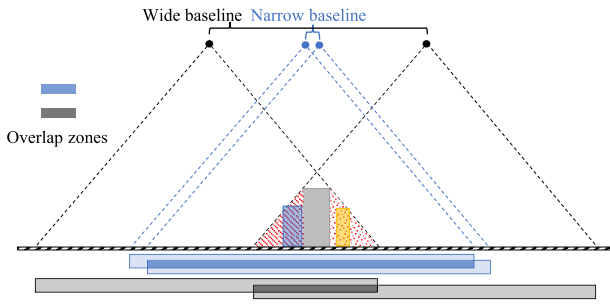


Fig. 2. Wide baseline and narrow baseline stereo configuration. In the wide baseline stereo configuration, the occlusions are signaled by the red areas, and the small blue and yellow buildings are almost completely occluded by the gray taller building. The blue and dark gray rectangles represent the usable overlapping areas of narrow and wide baseline stereo configurations, respectively.

can be improved by using a system with a relatively wide baseline.

Stereo systems with narrow baseline generally lack the accuracy necessary for disparity measurements, and hence a wider baseline is usually used. However, due to the large scene variation and the small overlapping area, a wider baseline introduces other issues, such as difficult matches and larger occlusions [5]. The processes for capturing wide and narrow baseline image pairs are compared in Fig. 2. The small blue and yellow buildings can hardly be captured simultaneously in the wide baseline stereo configuration, but they are visible in narrow one. The usable overlapping image areas increases as the baseline decreases.

In the case of a wide baseline, there are large occlusion areas and geometric distortions between images. In the case of the narrow baseline, there is a small occlusion area and limited geometric distortion between images. A very narrow baseline can minimize occlusion, and as a result, each image view will be more geometrically similar. Moreover, the interval between the two imaging processes becomes short, and the illumination conditions are more similar.

When $H \gg h$, according to (2), the disparity d is proportional to the B/H . Considering a certain image resolution, traditional aerospace photogrammetry adopts the scheme of a large base/height ratio (e.g., 0.6/1) to obtain high elevation accuracy, high-precision extraction of 3-D terrain information. Most stereo vision researches have focused on cases with large B/H ratio. However, when methods that use a large B/H ratio are applied in environments with high rates of fluctuation and drastic changes, they are likely to cause significant radiation and geometric differences between stereo images. These differences, affected by occlusion, distortion, and moving objects, make correlation matching processing extremely difficult. Under small B/H conditions, the image capturing time interval is very short, and the geometric distortion and radiation distortions become smaller. Although high matching accuracy is required, utilizing a large B/H ratio setting helps avoid many of its associated disadvantages.

When estimating depth from UAV consecutive frames, the baseline is generally narrow, and wide baseline stereo is usually not consistent with the situation. Therefore, we pay attention to the study of narrow one. Most stereo vision

algorithms can reliably calculate disparities with the integer-pixel precision. This may be adequate for certain applications, but not for narrow baseline stereo vision. In fact, when both views are separated by a very small angle, the disparities observed on images can be very small relative to the pixel size. In this case, the pixel-level precision matching method cannot detect any interesting depth information. Therefore, small baseline stereo vision requires matching methods for subpixel disparities [6].

Morgan et al. [4] developed a stereo matching algorithm based on the robust phase correlation method to directly measure disparities up to 1/50th pixel accuracy and precision when $B/H < 0.01$. Equations (3) and (4) are used to show that when a UAV flies high enough, two corresponding local patches in two consecutive frames satisfy a translation transformation.

Consider two image points, $\mathbf{x}_1 = (x_1, y_1)^T$ and $\mathbf{x}_2 = (x_2, y_2)^T$, corresponding to the same 3-D point $\mathbf{X} = (X, Y, Z, 1)^T$. K is called the camera calibration matrix. I and C are the identity matrix and camera center, respectively. Rotation is ignored because two consecutive frames are very close to each other. According to [7], the relationship between two image points can be expressed as

$$\begin{aligned} z_2 \cdot \begin{bmatrix} x_2 \\ y_2 \\ 1 \end{bmatrix} &= K \cdot [I \mid -C] \cdot \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} \\ &= K \cdot \left[K^{-1} \cdot \begin{bmatrix} x_1 \\ y_1 \\ 1 \end{bmatrix} \cdot z_1 \right] - KC \\ &= [x_1 \quad y_1 \quad 1]^T \cdot z_1 - KC. \end{aligned} \quad (3)$$

When the UAV flies high enough (i.e., when the height of the camera H is large enough), it is reasonable to assume that $H \gg h$, if we assume that the depth of the local patch is constant, the depths z_1 and z_2 are approximately the same, and (3) can be transformed into

$$z \left(\begin{bmatrix} x_2 \\ y_2 \\ 1 \end{bmatrix} - \begin{bmatrix} x_1 \\ y_1 \\ 1 \end{bmatrix} \right) = -KC. \quad (4)$$

Image registration must be used to select a region as individual pixels cannot be registered. In [4], the depth between two local patches in two continuous frames are regarded as consistent, and the relationship between the two image points is expressed by (4). In this case, two corresponding local patches in two consecutive frames satisfy a translation transformation, and image translation information can be directly calculated to determine the image disparity; thus, subpixel accuracy can be achieved by using the phase correlation.

As depicted in Fig. 1, regardless of the UAV's altitude and its orientation relative to the ground, geometric deformations persist between continuously captured images. However, the geometric transformations between images involve more than translation transformations, and therefore, the method in [4] based on translation transformations is no longer applicable. If the depth of the local patch is approximately planar, the local patches satisfy the affine transformation (the proof for this process is presented later in this article).

Herein, we propose a precise and robust depth estimation method based on narrow baseline: we give the mathematical proofs that two corresponding local patches in two consecutive frames satisfy an affine transformation, and provide solution for the affine transformation parameters. Additionally, we apply the Fourier moment matching method in [8] to deduce the linear part of transformation parameters by minimizing the affine relationship between the Fourier moments of two images, in particular, we introduce differential geometry to analyze the 3-D patch. This method reduces the influence of geometric deformation on depth estimations. Finally, the translation vector is calculated using SLSF-PC in [9], which can suppress the illumination influence. Upon solving the affine transformation parameters, we can calculate the normal of the 3-D patch to achieve depth estimation.

In summary, the main contributions of this article are threefold.

- 1) By introducing multiview geometry theory, we prove that the affine relationship is satisfied in two local patches between two consecutive frames no matter how high the UAV flies and whether it is parallel to the ground or not. Affine transformation parameters can be divided into two components: affine transformation matrix and translation vector.
- 2) A Fourier moment matching algorithm combined with the normal curvature (the degree of curvature of the image surface at a specific direction) is used to calculate the affine transformation matrix. The SLSF-PC method is applied to calculate the translation vector.
- 3) The normal and position of the 3-D patch corresponding to the local image patches in two consecutive frames can be directly calculated based on affine transformation parameters instead of finding the best fitting 3-D plane for each pixel in the image, and the depth of the image can be calculated according to the condition that the 3-D point is on the plane.

The remainder of this article is organized as follows. Related works are discussed in Section II. Details of the depth estimation method by calculating affine transformation parameters are described in Section III. The proposed method is verified based on experiments carried on datasets for real UAV images and real images of scale model scenes in Section IV. The conclusions are presented in Section V.

II. RELATED RESEARCH

The proposed method in this article is based on Fourier transform for image registration, aiming to achieve narrow baseline stereo matching and ultimately enable dense depth estimation. This work is related to various stereo vision matching methods, subpixel image registration methods, and precise depth estimation techniques. This section focuses on the methods that are closely related to the method applied in our research (within the scope of the three aforementioned topics).

A. Stereo Vision Matching Methods

The goal of stereo vision matching is to find the precise locations of corresponding points in two images. From the perspective of matching results, it can be divided into sparse

matching and dense matching. Sparse matching typically uses point features [such as scale-invariant feature transform (SIFT) [10], oriented fast and rotated brief (ORB) [11], features from accelerated segment test (FAST) [12], and so on] to obtain relatively accurate matching results between the two images. However, point features may fail in sparse or repetitive texture regions, making sparse matching unsuitable for high-precision and high-integrity 3-D reconstruction. When the time interval between image captures is small, i.e., the baseline is narrow, dense matching methods can be used to obtain dense disparity estimation, thus achieving more complete depth calculation results and enabling high-integrity 3-D reconstruction. In the PatchMatch algorithm proposed by Connelly Barnes et al. [13], global approximate optimal matching is achieved through the propagation of nearest-neighbor matching patches and iterative refinement. However, PatchMatch performs poorly in matching regions with weak textures and discontinuous disparities. The semi-global matching (SGM) algorithm proposed by Hirschmuller [14] uses mutual information for matching under the assumption of known epipolar geometry. It exhibits robustness against variations in lighting, noise, and depth changes.

Phase correlation method is a mainstream matching method. For example, Wan et al. [15] proposed to accurately estimate the position of UAVs by matching UAVs images with reference satellite images using phase-dependent light invariant characteristics. pei Zhao et al. [16] improved the multiscale phase method by replacing the gray level information in the traditional phase correlation method with the wavelet modulus value to improve the matching accuracy. Ahmed and Jafri [17] proposed a preprocessing technique to effectively deal with false peaks in phase correlation matching, which improved the robustness of the phase correlation method. Window-based phase-related stereo matching algorithm is a common local stereo matching algorithm. For example, Wenping and Changling [18] proposed an image matching correction method using window series integration, and used least squares and phase-related matching to correct the corresponding points in the window series. Sakai et al. [19] used phase correlation method and mostly geometric correction method for window matching, and the final matching result achieved subpixel accuracy. Fusiello et al. [20] propose an adaptive multiwindow scheme that uses left-right consistency to calculate parallax and solves the problem of robust parallax estimation in the case of occlusion. In the problem of 3-D reconstruction of UAV continuous image studied in this article, the shooting process of two images meets the premise that the baseline is narrow. The disadvantage of narrow baseline stereo methods is that higher accuracy is required when calculating the disparities [21]. In the case of narrow baselines, the information required for matching cannot be accurately extracted from the gray level of a single pixel, and therefore, image depth must be calculated based on subpixel precision registration.

B. Subpixel Image Registration Methods

At present, subpixel-level image registration algorithms include gray interpolation [22], based solution optimization [23], and based transform domain methods. Gehrig et al. [24]

proposed a subpixel accuracy method for long-distance binocular stereo reconstruction, which improves the accuracy by interpolating and concealing low-texture areas, but the method based on grayscale interpolation has statistical bias and low registration accuracy. The image registration method based on Fourier has high computational efficiency and provides anti-noise ability, which has been improved by many researchers. For example, Niu et al. [25] proposed a method based on the combination of Fourier transform and traditional phase correlation technology for image registration, which has superior noise suppression performance. The multilevel fast Fourier transform (MLFFT) developed by Pan et al. [26] has the same computational efficiency and higher accuracy as the pseudopolar fast Fourier transform (FFT). Zhang et al. [27] further expanded the method based on Fourier transform by solving mean square error (MSE) estimation of image registration parameters in the fractional Fourier transform domain. Morgan et al. [4] proposed an accurate subpixel parallax calculation method, which is effective for calculating dense parallax maps under narrow baselines. Takita et al. [28] proposed to use pyramid strategy to search for corresponding pixel registration, and found subpixel level parallax of relevant points in a coarse-to-fine way, and obtained high-quality parallax maps using small-window estimation. Li et al. [29] realized subpixel precision registration based on pyramid phase correlation and upsampling. Ye et al. [30] realized robust and fine registration based on enhanced subpixel phase correlation. Argyriou and Tzimiropoulos [31] used histogram of oriented gradient (HOG) phase correlation to achieve subpixel precision registration in the frequency domain. Ren et al. [32] realized high-precision subpixel-level motion estimation in noisy images based on Fourier domain phase correlation method, showing the noise suppression performance of Fourier domain phase correlation method.

It is also necessary to consider how to solve the problem of illumination invariant matching. Tan et al. [33] developed an illumination correction algorithm based on an empirical rotation model, where the illumination correction process successfully eliminated correlations between Landsat reflectance and the illumination conditions. The digital elevation model contains stereo matching errors, which significantly affect the correction results. Wan et al. [9] proposed a model derived from phase correlation decomposition (PCD) theory to analyze the effects of zenith and azimuth angles of the lighting source; they also developed a SLSF-PC method, which suppressed the illumination effect significantly, and this method is adopted in our present work.

C. Depth Estimation Methods

For UAV 3-D reconstruction, including active LiDAR and structured light schemes, but limited by equipment accuracy, applicable scenarios and cost ratio, passive image-based 3-D reconstruction methods are more common and widely used. Dense depth estimation is the upstream task of 3-D reconstruction, and dense depth estimation for UAV images has been widely studied. The traditional image dense depth estimation method is based on the mature multiview geometry mathematical model, and adopts hand-designed features for

camera pose estimation and dense depth propagation. For example, Wang et al. [34] has developed a set of effective methods to find the correct depth value of the texture region, which is usually difficult to retain on the fine scale. Xu and Tao [35], [36], [37] proposed multiscale geometric consistency guided multiple view stereo (MVS) methods, adaptive checkerboard sampling and multihypothesis joint view selection (ACMH), multiscale geometric consistency guidance framework (ACMM), and planar prior assisted framework (ACMP), which can effectively recover the depth information of weak texture regions. Zheng et al. [38] proposed a multiview depth map estimation approach that jointly models pixel-level view selection and depth map estimation given the local pairwise image photoconsistency. Shen [39] proposed an efficient patch-based stereo matching process that generates a depth map at each image with acceptable errors, followed by a depth map refinement process to enforce consistency over neighboring views.

Wu et al. [40] proposed a geometry guided multiscale depth map fusion method, considering both the sampling scale of a point and relations among points, which can reduce the noise created by multiple view stereopsis. Based on the Patchmatch method, Galliani et al. [41] started from randomly generated 3-D planes in a scene space. For the depth estimation of UAV continuous frames, the traditional depth estimation method based on binocular reconstruction has developed a complete mathematical model and mature solution, which can achieve better results. The reconstruction algorithms based on multiview geometry are usually based on the iterative optimization of PatchMatch to estimate the depth, but such methods are plagued by weak texture regions. The method proposed in this article is based on the minimum iterative solution of Fourier moments and directly calculates the parameter information of the plane rather than initializing the random plane for each pixel, which has higher matching accuracy.

III. DEPTH ESTIMATION SCHEME BY CALCULATING AFFINE TRANSFORMATION PARAMETERS

The depth estimation scheme by calculating affine transformation parameters is described in detail in this section. Fig. 3 succinctly describes the geometry of the problem and provides a detailed description of the process.

A. Affine Relationships in Local Regions

Traditional stereo matching methods typically use similar exhaustive methods to find matching image points. For example, patch-based methods represent scene surfaces by collections of small patches, and they assign a random depth z and a random normal $\mathbf{n}$ for every pixel in the reference image; these parameters are used to compute the optimal tangent plane of each pixel onto which the support region is projected by finding a depth z and a normal $\mathbf{n}$ that minimize the matching cost between the reference image and its neighboring images. Bleyer et al. [42] introduced PatchMatch into binocular stereo vision, which enabled the identification of an approximate optimal tangent plane of each pixel in the

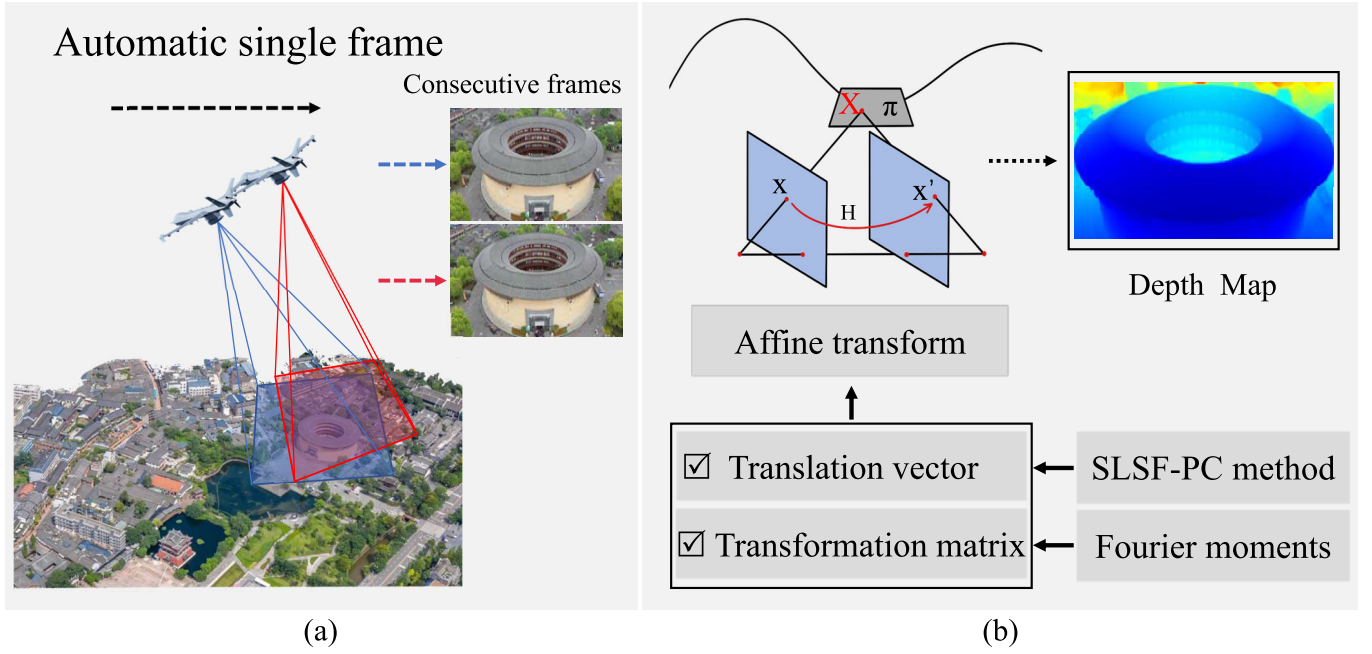


Fig. 3. Complete flow diagram of the depth estimation scheme by calculating affine transformation parameters. (a) shows a set of consecutive frames captured from the UAV. In (b), along the direction of the arrow, the linear part of affine transformation parameters is obtained using Fourier moments. Next, the translation vector of the affine transformation parameters is determined using the SLSF-PC method. Finally, the image depth information is derived.

disparity space images (DSIs) from corrected image pairs. In contrast, we directly calculate the normal of 3-D patch in the present work. In terms of the depth z , different from existing methods that select a random depth within the range of allowed continuous depth values, we compute z using the method described hereafter.

Considering the projection matrices for the two views $P = [I | \mathbf{0}]$ and $P' = [A | \mathbf{a}]$, matrix I is a 3×3 identity matrix, $\mathbf{0}$ is a zero 3-vector, A is a 3×3 matrix corresponding to the second projection matrix, and vector $\mathbf{a}$ is the epipole in second view arising from the first camera. According to [7], point $\mathbf{x}$ in the first view is backprojected such that the intersection point $\mathbf{X}$ of this ray with the plane π is determined. Thus, $\mathbf{x} = P\mathbf{X} = [I | \mathbf{0}]\mathbf{X}$, and any point on the ray $\mathbf{X} = (\mathbf{x}^T, \rho)^T$ projects to $\mathbf{x}$, where ρ parametrizes the point on the ray. Because the 3-D point $\mathbf{X}$ is on the plane $\pi = (\mathbf{v}^T, 1)^T$, it satisfies $\pi^T \mathbf{X} = 0$, and the vector $\mathbf{v}$ specifies a three-parameter of plane π , thus, ρ is determined, and $\mathbf{X} = (\mathbf{x}^T, -\mathbf{v}^T \mathbf{x})^T$. Then, the 3-D point $\mathbf{X}$ is projected onto the second view, and the projection point $\mathbf{x}' = P'\mathbf{X} = [A | \mathbf{a}]\mathbf{X} = (A - \mathbf{a}\mathbf{v}^T)\mathbf{x}$. The homography H induced by the plane π is $\mathbf{x}' = H\mathbf{x}$ with

$$H = A - \mathbf{a}\mathbf{v}^T. \quad (5)$$

We suppose that the world origin is at the first camera center, and the camera matrices are those of a calibrated stereo rig $P_E = K_E[I | \mathbf{0}]$ and $P_{E'} = K_{E'}[R_E | \mathbf{t}_E]$. The plane π_E has coordinates $\pi_E = (\mathbf{n}^T, \pi_4)^T$. According to (5), with $\mathbf{v} = \mathbf{n}/\pi_4$, the homography for the cameras $\tilde{P}_E = [I | \mathbf{0}]$ and $\tilde{P}_{E'} = [R_E | \mathbf{t}_E]$ is

$$\tilde{H}_E = R_E - \mathbf{t}_E \mathbf{n}^T / \pi_4. \quad (6)$$

By applying the transformations K_E and $K_{E'}$ to the images, the resulting induced homography is

$$H_E = K_{E'}(R_E - \mathbf{t}_E \mathbf{n}^T / \pi_4) K_E^{-1}. \quad (7)$$

Each pixel in the image has a local tangent plane corresponding to 3-D points in the space. The local tangent plane of the scene surface is represented by the 3-D coordinates of the point $\mathbf{X}$ and its normal $\mathbf{n}$. In our method, let $\mathbf{n} = [\alpha, \beta, \gamma]^T$ and let $\mathbf{X} = [X, Y, Z]^T$. In the camera coordinate system of the reference image, we define the local tangent plane π_I has coordinates $\pi_I = (\mathbf{n}^T, \pi_4)^T$, and for points on the local tangent plane

$$\alpha X + \beta Y + \gamma Z + \pi_4 = 0. \quad (8)$$

For a pair of images I and I' in the image set, we back-project a point $\mathbf{x}$ on the first image I and determine the intersection point $\mathbf{X}$ of this ray with the π_I . The 3-D point $\mathbf{X}$ is then projected onto the second image I' so that the point $\mathbf{x}'$ is obtained. The associated camera parameters of the image I are the intrinsic parameters K_I , the rotation matrix R_I , and the camera center C_I . The associated camera parameters of the image I' are the intrinsic parameters $K_{I'}$, the rotation matrix $R_{I'}$, and the camera center $C_{I'}$. Thus, $\mathbf{t}_I = C_I - C_{I'}$, according to (6), the projection matrices and plane parameters can be translated into standard forms, the homography for the cameras $\tilde{P}_I = [I | \mathbf{0}]$ and $\tilde{P}_{I'} = [R_{I'} R_I^{-1} | R_{I'}(C_I - C_{I'})]$ is

$$\tilde{H}_I = R_{I'} R_I^{-1} - \frac{R_{I'}(C_I - C_{I'}) \mathbf{n}^T}{\pi_4}. \quad (9)$$

By applying the transformations K_I and $K_{I'}$ to the images, we obtain the cameras $P_I = K_I[I | \mathbf{0}]$, $P_{I'} = K_{I'}[R_{I'} R_I^{-1} |$

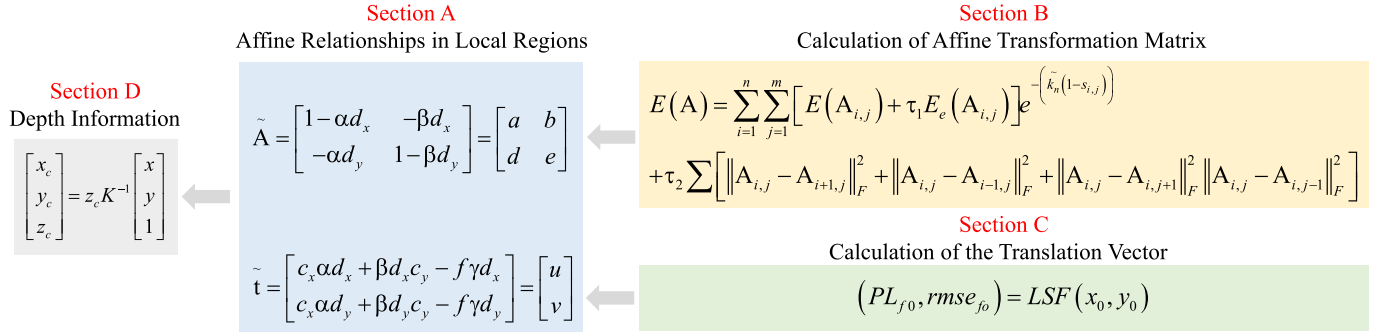


Fig. 4. Diagram illustrating the relationship between formulas in different sections.

$R_{I'}(C_I - C_{I'})$] and the resulting induced homography is

$$\begin{aligned}
 H_I &= K_{I'} \left(R_{I'} R_I^{-1} - \frac{R_{I'}(C_I - C_{I'}) \mathbf{n}^T}{\pi_4} \right) K_I^{-1} \\
 &= \begin{bmatrix} f & 0 & c_x \\ 0 & f & c_y \\ 0 & 0 & 1 \end{bmatrix} \left(\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \frac{\begin{bmatrix} d_x \\ d_y \\ 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix}^T}{1} \right) \\
 &\quad \cdot \begin{bmatrix} \frac{1}{f} & 0 & \frac{-c_x}{f} \\ 0 & \frac{1}{f} & \frac{-c_y}{f} \\ 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 1 - d_x \alpha & -d_x \beta & c_x d_x \alpha + c_y d_x \beta - f d_x \gamma \\ -d_y \alpha & 1 - d_y \beta & c_x d_y \alpha + c_y d_y \beta - f d_y \gamma \\ 0 & 0 & 1 \end{bmatrix} \quad (10)
 \end{aligned}$$

where $C_I = [0, 0, 0]^T$, $C_{I'} = [d_x, d_y, 0]^T$, c_x and c_y represent the coordinates of the principal point on the image plane. We assume that $\pi_4 = 1$ because the π_I does not pass through the center of the first camera C_I . Considering that two consecutive frames are very close to one another, the rotation is ignored. Thus, the rotation matrices R_I and $R_{I'}$ are 3×3 identity matrices. When the cameras of two consecutive frames are the same, the intrinsic parameters K_I and $K_{I'}$ are the same, let

$$K_I = K_{I'} = \begin{bmatrix} f & 0 & c_x \\ 0 & f & c_y \\ 0 & 0 & 1 \end{bmatrix} \text{ and } K_I^{-1} = K_{I'}^{-1} = \begin{bmatrix} \frac{1}{f} & 0 & \frac{-c_x}{f} \\ 0 & \frac{1}{f} & \frac{-c_y}{f} \\ 0 & 0 & 1 \end{bmatrix}.$$

The image 2-D points $\mathbf{x}$ and $\mathbf{x}'$ are corresponded to the same 3-D point $\mathbf{X}$, denote $[x_1 \ y_1 \ 1]^T$ the homogeneous coordinates of the 2-D point $\mathbf{x}$ and $[x_2 \ y_2 \ 1]^T$ the homogeneous coordinates of the 2-D point $\mathbf{x}'$. Then, the homography induced by the local tangent plane is

$$\begin{bmatrix} x_1 & y_1 & 1 \end{bmatrix}^T = H \begin{bmatrix} x_2 & y_2 & 1 \end{bmatrix}^T. \quad (11)$$

In order to obtain the relationships between two corresponding local patches in two consecutive frames, we combine (10) and (11) as follows:

$$\begin{bmatrix} x_2 \\ y_2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 - \alpha d_x & -\beta d_x & c_x \alpha d_x + \beta d_x c_y - f \gamma d_x \\ -\alpha d_y & 1 - \beta d_y & c_x \alpha d_y + \beta d_y c_y - f \gamma d_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ 1 \end{bmatrix}. \quad (12)$$

We can simplify (12) as follows:

$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 - \alpha d_x - \beta d_x \\ -\alpha d_y - \beta d_y \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} + \begin{bmatrix} c_x \alpha d_x + \beta d_x c_y - f \gamma d_x \\ c_x \alpha d_y + \beta d_y c_y - f \gamma d_y \end{bmatrix}. \quad (13)$$

Substitute the affine transformation matrix and the translation vector into (13)

$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \tilde{A} * \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} + \tilde{\mathbf{t}}. \quad (14)$$

Therefore, there is an affine transformation relationship between two 2-D points. By solving the affine transformation matrix $\tilde{A} = \begin{bmatrix} 1 - \alpha d_x & -\beta d_x \\ -\alpha d_y & 1 - \beta d_y \end{bmatrix}$ (as described in Section B) and the translation vector $\tilde{\mathbf{t}} = \begin{bmatrix} c_x \alpha d_x + \beta d_x c_y - f \gamma d_x \\ c_x \alpha d_y + \beta d_y c_y - f \gamma d_y \end{bmatrix} = \begin{bmatrix} u \\ v \end{bmatrix}$ (as described in Section C), it is possible to calculate the depth for the image (as described in Section D). Fig. 4 shows the relationship between equations from different sections.

B. Calculation of Affine Transformation Matrix

1) *Affine Relations of Fourier Spectrums*: Consider two images $f_1(x, y)$ and $f_2(x, y)$, $x, y \in \mathbf{R}^2$, which are related by the following affine transformation

$$f_1(x, y) = f_2(ax + by + u, d x + ey + v). \quad (15)$$

We define the Fourier transforms of the image functions $f_1(x, y)$ and $f_2(x, y)$ as $F_1(h, k)$ and $F_2(h, k)$, respectively. Then, according to [43], we derive the affine relation between their Fourier transforms $F_1(h, k)$ and $F_2(h, k)$ by the following formulas.

We express the affine coordinate transformation of two images in matrix notation

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a & b \\ d & e \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} u \\ v \end{bmatrix}. \quad (16)$$

The linear part of the affine transformation in (16), Δ is defined as

$$\Delta = \begin{bmatrix} a & b \\ d & e \end{bmatrix}, \quad dx'dy' = |\Delta|dxdy. \quad (17)$$

In the phase exponent $-i2\pi(hx + ky)$, the 2-D Fourier component is defined as

$$\begin{aligned} hx + ky &= \begin{bmatrix} h & k \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \\ &= \begin{bmatrix} h & k \end{bmatrix} \begin{bmatrix} a & b \\ d & e \end{bmatrix}^{-1} \begin{bmatrix} x' - u \\ y' - v \end{bmatrix} \\ &= \frac{1}{\Delta} \begin{bmatrix} eh - dk & -bh + ak \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} \\ &\quad - \frac{1}{\Delta} \begin{bmatrix} eh - dk & -bh + ak \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix}. \end{aligned} \quad (18)$$

Thus, we can obtain the following affine transformation relationship between the Fourier transforms $F_1(h, k)$ and $F_2(h, k)$

$$\begin{aligned} F_1(h, k) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_2(x', y') \times e^{-2\pi i(hx + ky)} dx dy \\ &= e^{i(2\pi/\Delta)[(ec-bf)h + (af-cd)k]} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_2(x', y') \\ &\quad \times e^{-i(2\pi/\Delta)[(eh-dk)x' + (-bh+ak)y']} dx' dy' / |\Delta| \\ &= \frac{1}{|\Delta|} e^{\frac{i2\pi[(ec-bf)h + (af-cd)k]}{\Delta}} F_2\left(\frac{eh-dk}{\Delta}, \frac{-bh+ak}{\Delta}\right). \end{aligned} \quad (19)$$

Taking the absolute values on both sides of (19), we have the affine transformation relationship between the spectrums $|F_1(h, k)|$ and $|F_2(h, k)|$ as follows

$$|F_1(h, k)| = \frac{1}{|\Delta|} |F_2(h', k')| \quad (20)$$

where $h' = (eh - dk)/\Delta$ and $k' = (-bh + ak)/\Delta$. We can know that if two images $f_1(x, y)$ and $f_2(x, y)$ are related by an affine transform, and their Fourier spectrums $F_1(h, k)$ and $F_2(h, k)$ are also related by the corresponding affine transform.

2) *Calculation Based on Fourier-Moment Matching:* The affine parameters can be solved by the Fourier moment matching technique [8]. The $(i + j)$ th moment for the Fourier spectrum $|F_n(h, k)|$ is defined as

$$m_{i,j}^n = \iint h^i k^j |F_n(h, k)| dh dk \quad (21)$$

where $n = 1$ or 2 , by coordinate substitution, we can derive the following equation:

$$m_{i,j}^2 = \iint (ah' + bk')^i (dh' + ek')^j |F_2(h', k')| dh' dk'. \quad (22)$$

According to [44], for the moment, we will ignore the translational component of affine transformation parameters and focus on its linear component in (16). Consider the linear part of affine transformation in which the linear monomials x and y are transformed into the following two linear monomials:

$$x' \rightarrow ax + b, \quad y' \rightarrow dx + ey, \quad A_1 = \begin{bmatrix} a & b \\ d & e \end{bmatrix} \quad (23)$$

where A_1 is the linear part of the affine transformation of the first-order moments. These can be extended to define transformations for monomials of higher degrees using polynomial multiplications. For example, the transformations for the following three monomials can be defined

$$\begin{aligned} x'^2 &\rightarrow (ax + by)^2 = a^2x^2 + 2abxy + b^2y^2 \\ x'y' &\rightarrow (ax + by)(dx + ey) = adx^2 + (ae + bd)xy + bey^2 \\ y'^2 &\rightarrow (dx + ey)^2 = d^2x^2 + 2dexy + e^2y^2. \end{aligned} \quad (24)$$

Based on the transformations for the three basis monomials, we can define a linear transformation A_2 for the space related to the quadratic polynomials shown above as

$$A_2 \equiv \begin{bmatrix} a^2 & 2ab & b^2 \\ ad & ae + bd & be \\ d^2 & 2de & e^2 \end{bmatrix} \quad (25)$$

where A_2 is the linear part of the affine transformation of the second-order moments. Then, with coordinate substitution, we can derive the following relationships for the first-order moments

$$M_1^1 = \begin{bmatrix} m_{1,0}^1 \\ m_{0,1}^1 \end{bmatrix}, \quad M_1^2 = \begin{bmatrix} m_{1,0}^2 \\ m_{0,1}^2 \end{bmatrix}, \quad M_1^1 = A_1 M_1^2. \quad (26)$$

Similarly, we can derive the following relationships for the second-order moments

$$M_2^1 = \begin{bmatrix} m_{2,0}^1 \\ m_{1,1}^1 \\ m_{0,2}^1 \end{bmatrix}, \quad M_2^2 = \begin{bmatrix} m_{2,0}^2 \\ m_{1,1}^2 \\ m_{0,2}^2 \end{bmatrix}, \quad M_2^1 = A_2 M_2^2. \quad (27)$$

The affine parameters (a, b, d, e) can be estimated by minimizing the total errors associated with the constraints in (26) and (27) in a least-squares estimation framework [44]

$$E(A) = \sum_{i=1}^2 w_i \|M_i^{(1)} - A_i M_i^{(2)}\|^2 \quad (28)$$

where i is the degree of moment, and w_i is the weight associated with the Fourier moment constraint for the i th degree. This method can be applied to obtain the affine matrix information for two corresponding local patches in two consecutive frames. However, the affine matrix parameters are not sufficient to solve the normal $\mathbf{n}$. Then, we obtain the transformed image, and the two images with only translation transformation can be obtained, and the subpixel translation can be calculated by the method described in Section C to solve the normal $\mathbf{n}$.

3) *Local Affine Transformation:* We divide the calculation of the affine transformation parameters into two steps by referring to the Fourier moment matching method in [8]. First, the edge image of the reference image is divided into a set of patches. The cross power spectrum between each patch and the target image is calculated to determine the region corresponding to each patch in the target image. The similarity measure score fm for patch $q(x)$ and the target image $r(x)$ is computed as

$$fm = \frac{\max \text{real}(\text{IFT}((Q'(u)R^*(u))/|Q'(u)R^*(u)|))}{\max \text{real}(\text{IFT}((R(u)R^*(u))/|R(u)R^*(u)|))} \quad (29)$$

where $q'(x) = q(Ax)$ and $Q'(u)$ is the Fourier spectrum of $q'(x)$. $R^*(u)$ is the conjugate Fourier spectrum of $r(x)$, and IFT signifies the inverse Fourier transform. The term real means the real part of the complex number in the Fourier domain and maxreal is its max value. Thus, we can obtain a rough initial transformation between two images for the subsequent refinement step. Second, the image is partitioned uniformly into M -by- N patches overlapping by 50%. We use differential geometry theory to analyze the 3-D patch for the coarse-to-fine local affine estimation. The normal curvature of 3-D patch in one direction can be regarded as the degree of curvature of the 3-D patch in that direction. We consider the normal curvature value as the basis for the partition of layered blocks for coarse-to-fine local affine estimation, and the normal curvature value can be calculated using differential geometry [8].

The vector equation corresponding to the image is $r(x, y) = (x, y, w(x, y))$, where the gray value corresponding to the image point (x, y) is $w(x, y)$. The first and second partial derivatives of the vector equation are as follows $r_x = (1, 0, (dw(x, y)/dx))$, $r_y = (0, 1, (dw(x, y)/dy))$; obviously, these two vectors are linearly independent for $r_x \times r_y \neq 0$. We define r as the 3-D patch, and we define $P = P(x, y)$ as the point on the 3-D patch where the coordinate parameter is (x, y) . Consider $x = x_1$, where $r(x_1, y)$ is a curve in space, and its tangent vector at $y = y_1$ is $r_y(x_1, y_1) = (dr/dy)(x_1, y_1)$. Since the parametric curve $r(x_1, y)$ falls on the 3-D patch, $r_y(x_1, y_1)$ is tangent to the 3-D patch at point $P_0 = P(x_1, y_1)$. Similarly, consider $y = y_1$, where $r(x, y_1)$ is a curve in space, and its tangent vector at P_0 is $r_x(x_1, y_1) = (dr/dx)(x_1, y_1)$. According to the definition of a 3-D patch, $r_x(x_1, y_1)$ and $r_y(x_1, y_1)$ are not collinear. Thus, the two vectors span a plane tangent to the 3-D patch at point P_0 , which is called the tangent plane of the 3-D patch at point P_0 such that $r_x(x_1, y_1)$ and $r_y(x_1, y_1)$ are base vectors to the 3-D patch at point P_0 . $n_r = r_x(x_1, y_1) \times r_y(x_1, y_1)$ is a vector orthogonal to both $r_x(x_1, y_1)$ and $r_y(x_1, y_1)$ such that it is a normal vector of the 3-D patch at point P_0 .

Any tangent vector v_r of the 3-D patch r can be represented in the following form $v_r = pr_x + qr_y$, the squared length of the tangent vector is $\langle v_r, v_r \rangle = p^2 \langle r_x, r_x \rangle + 2pq \langle r_x, r_y \rangle + q^2 \langle r_y, r_y \rangle$, where $E = \langle r_x, r_x \rangle$, $F = \langle r_x, r_y \rangle$, $G = \langle r_y, r_y \rangle$, $r_x = (1, 0, w_x)$, $r_y = (0, 1, w_y)$, $r_{xx} = (0, 0, w_{xx})$, $r_{yy} = (0, 0, w_{yy})$, $\sqrt{E}$ is the length of the tangent vector r_x , $\sqrt{G}$ is the length of the tangent vector r_y , and $(F/(\sqrt{E}G))$ is the cosine of the angle between r_x and r_y . We consider a curve $r(x(t), y(t))$ on the 3-D patch, whose tangent vector is $(dr(t)/dt) = r_x x'(t) + r_y y'(t)$. The arc length s of the curve in $t_1 < t < t_2$ is $s = \int_{t_1}^{t_2} |(dr(t)/dt)| dt = \int_{t_1}^{t_2} (E(dx/dt)^2 + 2F(dx/dt)(dy/dt) + G(dy/dt)^2)^{1/2} dt$, we take the derivative of both sides of this formula and square it to get the following formula $(ds/dt)^2 = E(dx/dt)^2 + 2F(dx/dt)(dy/dt) + G(dy/dt)^2$. Thus, a quadratic differential form can be obtained, which is the first basic form of the 3-D patch

$$I = ds^2 = E dx \cdot dx + 2F dx \cdot dy + G dy \cdot dy. \quad (30)$$

In order to calculate the method curvature of the 3-D patch, the second basic form of the 3-D patch should also be calculated. The unit normal vector is $\tilde{n}_r = ((r_x \times r_y)/|r_x \times r_y|)$, according to the definition of a normal vector, $\langle r_x, \tilde{n}_r \rangle = 0$, $\langle r_y, \tilde{n}_r \rangle = 0$, we take the partial derivative of these, we obtain $\langle r_{xx}, \tilde{n}_r \rangle + \langle r_x, n_x \rangle = 0$, $\langle r_{xy}, \tilde{n}_r \rangle + \langle r_x, n_y \rangle = 0$, $\langle r_{yx}, \tilde{n}_r \rangle + \langle r_y, n_x \rangle = 0$ and $\langle r_{yy}, \tilde{n}_r \rangle + \langle r_y, n_y \rangle = 0$, and we define the function $L = \langle r_{xx}, \tilde{n}_r \rangle = -\langle r_x, n_x \rangle$, $M = \langle r_{xy}, \tilde{n}_r \rangle = -\langle r_x, n_y \rangle = -\langle r_y, n_x \rangle$ and $N = \langle r_{yy}, \tilde{n}_r \rangle = -\langle r_y, n_y \rangle$. The second fundamental form is a quadratic differential form, which can be expressed as

$$\begin{aligned} \Pi &= -\langle dr, d\tilde{n}_r \rangle = -\langle r_x dx + r_y dy, n_x dx + n_y dy \rangle \\ &= L dx dx + 2M dx dy + N dy dy. \end{aligned} \quad (31)$$

The first basic form of the 3-D patch represents the square of the arc differential of the curve on the 3-D patch, and the second basic form reflects the shape information of the 3-D patch at a certain point. Next, we calculate the normal curvature of the 3-D patch. $(x(s) \text{ and } y(s))$ is a curve that crosses a point (x_1, y_1) on a local area of the image, and $x(0) = x_1$, $y(0) = y_1$, $r(s) = r(x(s), y(s))$ is a curve passing through P_0 on the 3-D patch where P_0 corresponds to the parameter $s = 0$. The term s is the arc length parameter of the curve $r(s)$. The unit tangent vector of $r(s)$ at point P_0 is $(dr/ds)|_{s=0} = (r_x(dx/ds) + r_y(dy/ds))|_{s=0}$. The curvature vector of the curve $r(s)$ at point P_0 is $(d^2r/ds^2)|_{s=0} = (r_x(d^2x/ds^2) + r_y(d^2y/ds^2) + r_{xx}(dx/ds)^2 + 2r_{xy}(dx/ds)(dy/ds) + r_{yy}(dy/ds)^2)|_{s=0}$. Thus, the curvature vector is no longer the tangent vector of the 3-D patch, and the curvature vector is divided into the tangential and normal parts of the 3-D patch. We consider the curvature vector along the normal part of the 3-D patch, and we define

$$\begin{aligned} k_n &= \left\langle \frac{d^2r}{ds^2}, \tilde{n}_r \right\rangle \\ &= \langle r_{xx}, \tilde{n}_r \rangle \left(\frac{dx}{ds} \right)^2 + 2\langle r_{xy}, \tilde{n}_r \rangle \frac{dx}{ds} \frac{dy}{ds} + \langle r_{yy}, \tilde{n}_r \rangle \left(\frac{dy}{ds} \right)^2 \\ &= L \left(\frac{dx}{ds} \right)^2 + 2M \frac{dx}{ds} \frac{dy}{ds} + N \left(\frac{dy}{ds} \right)^2. \end{aligned} \quad (32)$$

Here, k_n is only related to the tangent vector $((dx/ds), (dy/ds))$ at point P_0 and the coefficients of the second basic form of the 3-D patch; it is independent of the selection of curve $x = x(s)$, $y = y(s)$. The term k_n is the normal curvature of the 3-D patch along the unit tangent vector $((dx/ds), (dy/ds))|_{s=0}$ at point P_0 and is the quadratic form of the 3-D patch with respect to the unit tangent vector at point P_0 .

The normal curvature of a 3-D patch along one direction can be regarded as the bending degree of the 3-D patch along that direction. The larger the curvature k_n of a region, indicating a higher degree of curvature within an image area, the greater the differences in affine transformation parameters among the pixels within that region.

We define the direction vector of the line $n_l = (\xi_l, \eta_l)$, the parametric equation of the line is $x(t) = \xi_l t + l_1$, $y(t) = \eta_l t + l_2$. The orthogonal projection of the curve on the image plane is a line, and the tangent vector of the curve at point P_0 is $\omega_l = \xi_l r_x + \eta_l r_y$, and its unit tangent vector

is $(\omega_l/|\omega_l|) = (\xi_l/(|\omega_l|))r_x + (\eta_l/(|\omega_l|))r_y$. Substitute the unit normal vector into (32)

$$k_n\left(\frac{\omega_l}{|\omega_l|}\right) = \frac{1}{|\omega_l|^2}(L\xi_l^2 + 2M\xi_l\eta_l + N\eta_l^2). \quad (33)$$

We can obtain $|\omega_l|^2 = \langle \xi_l r_x + \eta_l r_y, \xi_l r_x + \eta_l r_y \rangle = E\xi_l^2 + 2F\xi_l\eta_l + G\eta_l^2$. Thus, the normal curvature of the 3-D patch along the tangent vector $\omega_l = \xi_l r_x + \eta_l r_y$ is defined as

$$k_n(\omega_l) = \frac{\Pi(\omega_l, \omega_l)}{I(\omega_l, \omega_l)} = \frac{L\xi_l^2 + 2M\xi_l\eta_l + N\eta_l^2}{E\xi_l^2 + 2F\xi_l\eta_l + G\eta_l^2}. \quad (34)$$

We can calculate the line direction from the center of the image patch to the center of the four adjacent patches, and then calculate the normal curvature of the center of the image patch along the four directions to solve the average normal curvature $\tilde{k}_n$. Thus, we combine the local Fourier moment constraints, the spatial smoothness constraints, and the estimation of the 3-D patch using differential geometry. For each image patch in the image, the local affine estimation can be obtained by minimizing the following energy function,

$$E(A) = \sum_{i=1}^n \sum_{j=1}^m [E(A_{i,j}) + \tau_1 E_e(A_{i,j})] e^{-\tilde{k}_n(1-s_{i,j}))} + \tau_2 \sum \left[\|A_{i,j} - A_{i+1,j}\|_F^2 + \|A_{i,j} - A_{i-1,j}\|_F^2 + \|A_{i,j} - A_{i,j+1}\|_F^2 + \|A_{i,j} - A_{i,j-1}\|_F^2 \right] \quad (35)$$

where τ_1 is the weighting factor for the edge constraint, $E_e(A_{i,j})$ is the Fourier moment cost function for the (i, j) th block in the corresponding edge image, and $s_{i,j}$ is the fm value showing similarity of the corresponding local affine blocks. τ_2 is the weight for the smoothness constraint, and F is the Frobenius norm.

C. Calculation of the Translation Vector

The remaining translation vector $\tilde{\mathbf{t}}$ can be solved by stepwise least-squares fitting-based PC (SLSF-PC) [9]. The PC is based on the Fourier shift theorem. After calculating the affine transformation matrix $\tilde{A}$, we can obtain the transformed image $f'_2(x, y)$. Denote the translation vector $\tilde{\mathbf{t}} = (u, v)^T$. The relationship between the transform image and the reference image is $f_1(x, y) = f'_2(x + u, y + v)$, their Fourier transforms (FTs) [45] are $F_1(h, k)$ and $F'_2(h, k)$. The PC is defined by the normalized cross-power spectrum $Q(h, k)$ between the FTs $F_1(h, k)$ and $F'_2(h, k)$ of the two images. $Q(h, k) = (F_1(h, k)F'_2(h, k)^*)/(|F_1(h, k)F'_2(h, k)|) = e^{-i(uh+vk)}$ where $*$ stands for complex conjugate. There is a pixel-to-pixel multiplication between $F_1(h, k)$ and $F'_2(h, k)$.

Meanwhile, define an image $f_{\text{illumi}}(x, y)$ with only illumination difference from image $f_1(x, y)$, and its FT is $F_{\text{illumi}}(h, k)$. The illumination impact matrix Q_1 of $F_1(h, k)$ and $F_{\text{illumi}}(h, k)$ is $Q_1(h, k) = (F_1(h, k)F_{\text{illumi}}^*(h, k))/(|F_1(h, k)F_{\text{illumi}}(h, k)|)$, and define an image $f_{\text{shift}}(x, y)$ with only translation difference from image $f_1(x, y)$, and its FT is $F_{\text{shift}}(h, k)$, the translation matrix Q_2 of $F_1(h, k)$ and $F_{\text{shift}}(h, k)$ resulting from the 2-D shift between the two images is $Q_2(h, k) = e^{-i(uh+vk)}$. The PC

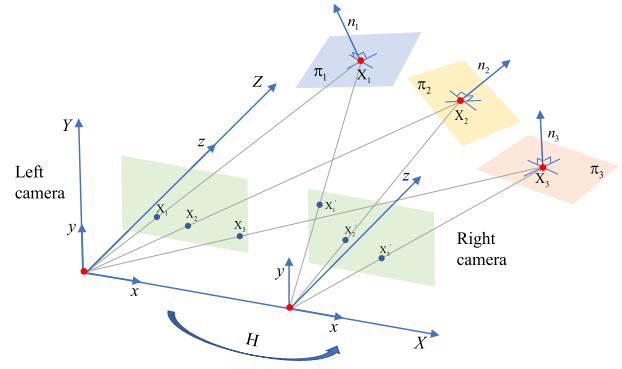


Fig. 5. Illustration of calculating the normals of local tangent planes corresponding to all pixels of the reference image. The image taken by the left camera is the reference image. The image points x_1 , x_2 , and x_3 are matched with x'_1 , x'_2 , and x'_3 on the image taken by the camera on the right, respectively, and correspond to π_1 , π_2 , and π_3 planes, respectively. n_1 , n_2 , and n_3 are the normals of the local tangent plane to be calculated, and X_1 , X_2 , and X_3 are the 3-D points corresponding to the image points.

matrix Q_3 between two images of $F_{\text{shift}}(h, k)$ and $F_{\text{illumi}}(h, k)$ which contains both illumination difference and image shift is $Q_3(h, k) = Q_1(h, k)Q_2(h, k)$. Q_3 can be calculated directly by phase correlation, and Q_1 and Q_2 are both unknown. By focusing on the joint effects of the variation illumination azimuth angle τ and zenith angle σ to PC, the translation matrix Q_2 can be extracted from the PC matrix Q_3 [9]. Thus, we can derive a more accurate translation vector and suppress the influence of illumination difference.

PC matrix Q_3 contains the positive and negative fringe patterns, can be denoted by $Q_P = e^{-i(uh+vk)}$ and $Q_N = -e^{-i(uh+vk)}$, respectively. In the case of different azimuth and zenith angles, the dominant singular vector will change in real time. SLSF-PC method automatically fits a single dominant singular vector in a direction for LSF. The LSF begins using 30% points from the dominant singular vector data in the initial fitting PL_{f0}

$$(\text{PL}_{f0}, \text{rmse}_{f0}) = \text{LSF}(x_0, y_0) \quad (36)$$

where LSF is the least-squares fitting function, x_0 and y_0 are fitting variables from the dominant singular vector in SVD [21], PL_{f0} and rmse_{f0} are the result function and root MSE (RMSE) from the fitting. Then, we gradually increase the numbers of points for fitting, as

$$(\text{PL}_{fn}, \text{rmse}_{fn}) = \text{LSF}(x_n, y_n), x_n = x_0 + n\Delta x \quad (37)$$

where $\Delta x = 0.1$. The LSF fitting will end when the RMSE exceeds a given threshold TLSF (TLSF = 0.1).

D. Calculation of the Depth Information

To compute the depth of a given pixel in the reference image, many existing methods follow the PatchMatch stereo methodology. These methods consider how to find an optimal patch match for each pixel by minimizing the matching cost, however, the proposed method directly calculates the parameter information of each patch. As shown in Fig. 5, we gradually calculate the normals of local tangent planes corresponding to all the pixels of the reference image.

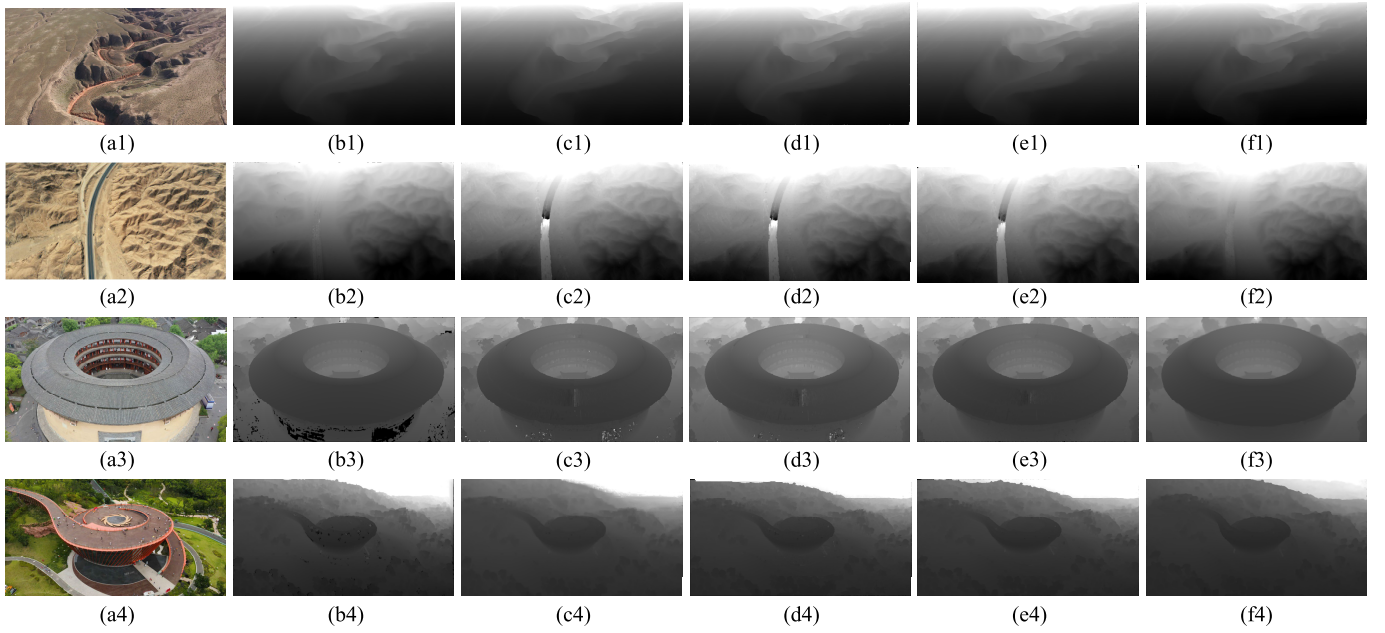


Fig. 6. Comparison of the depth maps obtained using various depth estimation methods. From left to right: (a1)–(a4) are the real images captured from the UAV; (b1)–(b4) are depth results from method [47]; (c1)–(c4) are depth results from the ACMH method; (d1)–(d4) are depth results from the ACMP method; (e1)–(e4) are depth results from the ACMMP method; (f1)–(f4) are depth results from our method.

The calculations described in Sections B and C can be used to solve for $\{\alpha, \beta, \gamma\}$, and then α , β , and γ can be calculated by

$$\begin{aligned} \alpha &= \frac{-d}{\sqrt{d^2 + (1-e)^2 + [(u+c_x a - c_x + c_y b)(1-e)]^2 / f^2 b^2}} \\ \beta &= \frac{1-e}{\sqrt{d^2 + (1-e)^2 + [(u+c_x a - c_x + c_y b)(1-e)]^2 / f^2 b^2}} \\ \gamma &= \frac{(u+c_x a - c_x + c_y b)(1-e)}{f b \sqrt{d^2 + (1-e)^2 + [(u+c_x a - c_x + c_y b)(1-e)]^2 / f^2 b^2}}. \end{aligned} \quad (38)$$

However, we assume that $\pi_4 = 1$ in Section A, the normal $\tilde{\mathbf{n}}$ of the π_I should be $\{\alpha/\tilde{\pi}_4, \beta/\tilde{\pi}_4, \gamma/\tilde{\pi}_4\}$. According to the length of the unit normal is 1, and therefore, the normal $\tilde{\mathbf{n}} = [n_1, n_2, n_3]^T$ can be computed, and the π_I can be expressed as $n_1 x_c + n_2 y_c + n_3 z_c + \tilde{\pi}_4 = 0$. Because the camera center of the first frame is set as the origin of the world coordinate system, the position of the image point in 3-D space can be expressed as

$$\begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix} = z_c K^{-1} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}. \quad (39)$$

The 3-D points (x_c, y_c, z_c) are located on the π_I , and therefore, the depth information z_c can be calculated.

IV. EXPERIMENTAL RESULTS

In this section, we evaluate our depth estimation scheme and compare it with various state-of-the-art methods, including spatial domain methods and a frequency domain method. We use real UAV images to verify the experimental results and

conduct a qualitative performance analysis. We also use real images of scale model scenes and compare the experimental results with other methods to give a quantitative analysis. All of the experiments described herein were conducted on an instrument equipped with Intel i7-9700K CPU and RTX 2080 GPU. Real UAV images were captured using a Canon EOS 5D Mark II digital single-lens reflex camera on a fixed-wing unmanned aerial vehicle (UAV), and real images of the scale model scenes were captured using a Hikvision MV-CH120-10UM industrial camera.

A. Real UAV Images

In Fig. 6(a1), (a2), (a3), and (a4), for each pair of these UAV images, the baseline direction is parallel to the image plane of the camera, and the baseline lengths are, respectively, about 3.8, 2.5, 1.8, and 2.1 m, and the camera look angles are, respectively, about 28°, 67°, 46°, and 59° from the ground, and the distances from camera to the image center are all almost 24 mm. To evaluate the performances of the selected depth estimation methods, we tested five methods on four UAV image datasets (Fig. 6). The condition of the narrow baseline stereo matching method [47] is that the overlap region of the UAV image pair is more than 82%. The constant C in the threshold function of method [47] is an empirical value, according to the description of the parameter in their article, we set this parameter as 4. ACMH [35] is a simple yet effective PatchMatch MVS method, which is the base model of ACMM [36] and ACMP [37]. A multiscale geometric consistency guided and planar prior assisted multiview stereo (ACMMP) [46] is an extended version of ACMM and ACMP. We converted the Structure-from-Motion (SfM) results in [48] to the input of ACMH, ACMP, and ACMMP.

Fig. 6 compares qualitative depth maps obtained using the developed method and other methods. Image Fig. 6(a1)

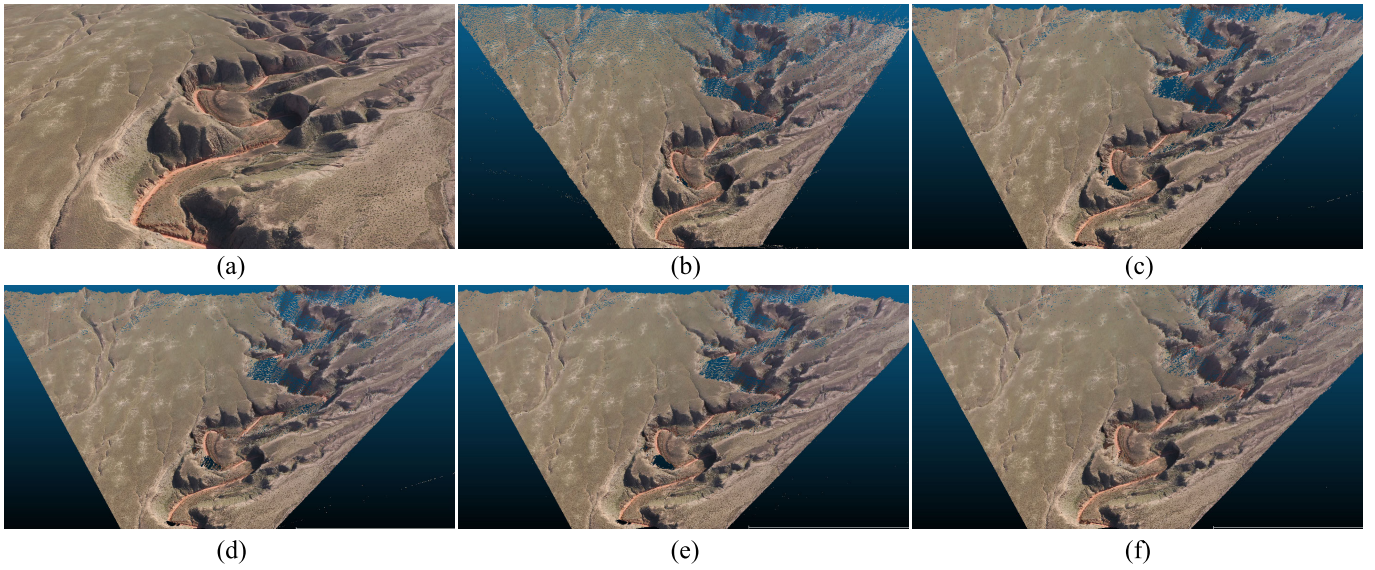


Fig. 7. Comparison of 3-D visualization results for the various methods in Fig. 6. (a) is image Fig. 6(a1); (b) is the result from method [47]; (c) is the result from the ACMH method; (d) is the result from the ACMP method; (e) is the result from the ACMMP method; and (f) is the result from our method.

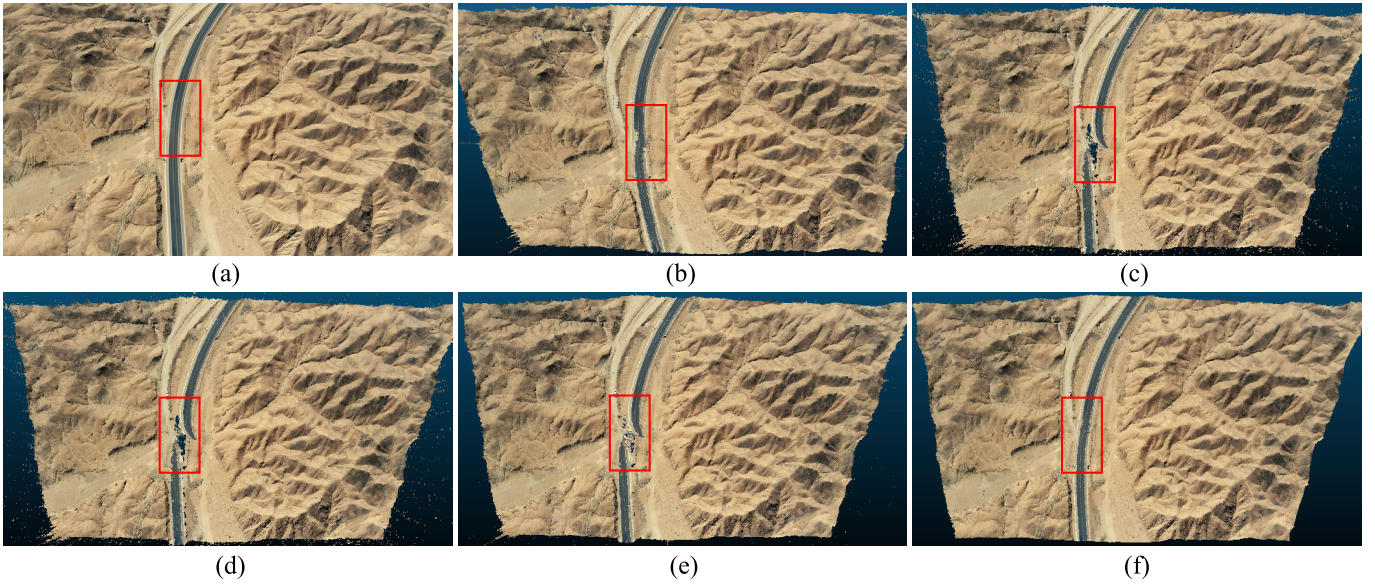


Fig. 8. Comparison of 3-D visualization results for the various methods in Fig. 6. (a) is image Fig. 6(a2); (b) is the result from method [47]; (c) is the result from the ACMH method; (d) is the result from the ACMP method; (e) is the result from the ACMMP method; and (f) is the result from our method.

shows the sinuous dry riverway. Comparing Figs. 6(b1)–(f1) and 7(b)–(f), it can be seen that the depth map produced by our method has more accurate results because the depth calculation in local areas is more complete than that in other methods. Image Fig. 6(a2) is the UAV image of a mountain country road. Figs. 6(c2)–(e2) and 8(c)–(e) show that the ACMH method, ACMP method, and ACMMP method have a large deviation in the depth estimation of low-texture areas. However, as shown by Figs. 6(b2) and (f2) and 8(b) and (f), method [47] and our developed method can obtain better results. Image Fig. 6(a3) shows the luodai ancient town in Chengdu. The regions with significant differences between our method and other methods are highlighted with red rectangles, as shown qualitatively in Fig. 6(b3). It is clear that the developed method achieves high-quality depth estimation in

complex structures and smooth regions, whereas the depth map produced by method [47] has more noise in low-texture areas. Image Fig. 6(a4) shows the longquan mountain dam in Chengdu. The developed method can produce a more complete depth map in comparison relative to method [47], ACMH method, ACMP method, and ACMMP method.

B. Real Images of Scale Model Scenes

In this section, two-scale model scenes (with buildings and a steep mountain, respectively) were constructed to quantitatively evaluate and compare the depth estimation methods. Each scene was photographed using a Hikvision MV-CH120-10UM industrial camera, which can give high-resolution images (4000×3060 pixels). Phase shift structured light technology was used to determine the ground

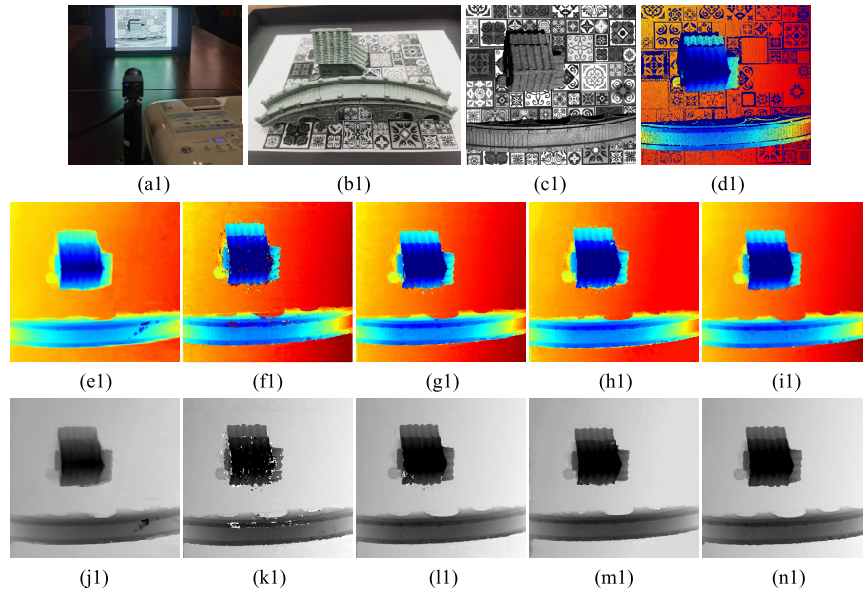


Fig. 9. Experimental results for the real images of buildings in a scale model scene obtained using the reference methods and our method. (a1) is a scene diagram of the structural camera image capturing process; (b1) is the real image captured by the structural camera; (c1) is the image of the scale model landscape scene for comparison; (d1) is the ground truth obtained using the phase shift structured light technology; (e1) and (j1) are the results from method [47]; (f1) and (k1) are the results from the ACMH method; (g1) and (l1) are the results from the ACMP method; (h1) and (m1) are the results from the ACMMP method; and (i1) and (n1) are the result from our method.

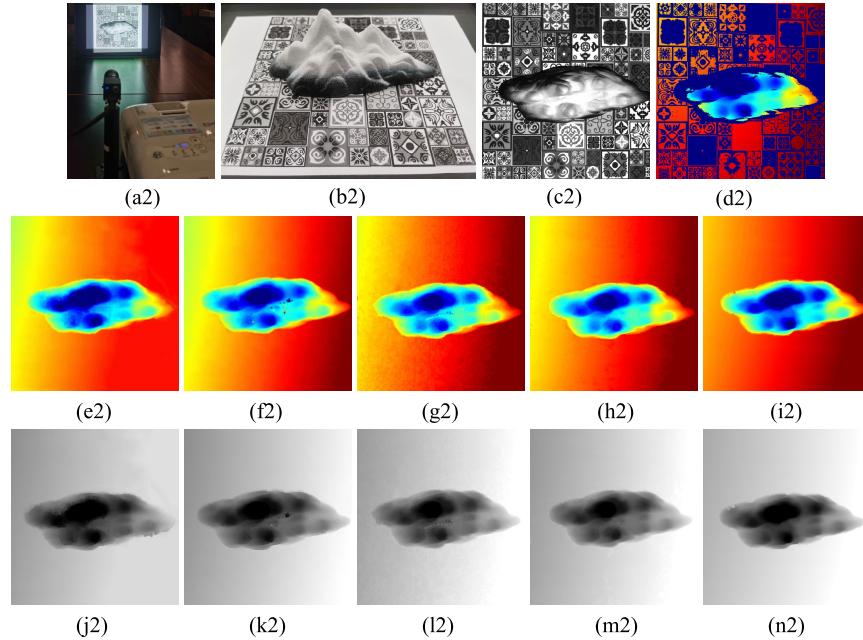


Fig. 10. Comparison of the depth maps corresponding to the real images of a steep mountain scale model scene. The annotations of images (a2), (b2), (c2), (d2), (e2), (f2), (g2), (h2), (i2), (j2), (k2), (l2), (m2), and (n2) are the same as those in Fig. 9.

truth of each image; these were designed in advance with special structure designs and then projected as 3-D objects on the surface, as shown in Fig. 9(a1). The projector was regarded as another camera, and we observed the 3-D physical surface distortion situation. The relationship between the pixels and the projector was elucidated according to the solution phase comprising the pixels and projector. Then, the principle of triangulation was applied to obtain the information related to 3-D points and overall depth information. Because the depth values calculated via phase shift structured light technology

are fairly accurate, we took the results as the ground truth.

1) Model of Buildings: The first scene contains two adjacent model buildings, where the precise heights of the house and the bridge were 70 and 50 mm, respectively, relative to the base. We imaged the scenes from a vertical height of 800 mm and moved the camera at a baseline length of 5 mm; the B/H ratio was 0.00625.

A visual comparison of the results obtained by various methods is shown in Fig. 9. The method developed in this work

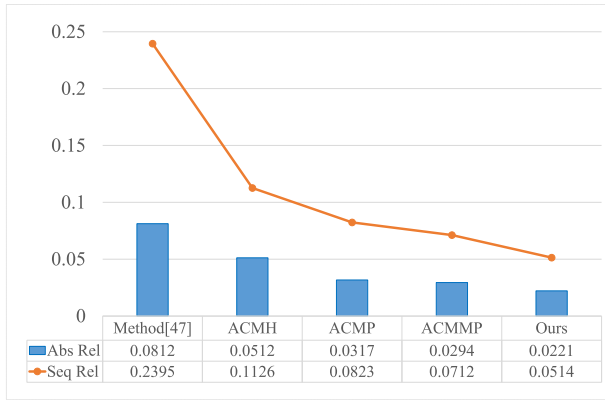


Fig. 11. Quantitative results from evaluation for the real images of buildings in a scale model scene (lower is better).

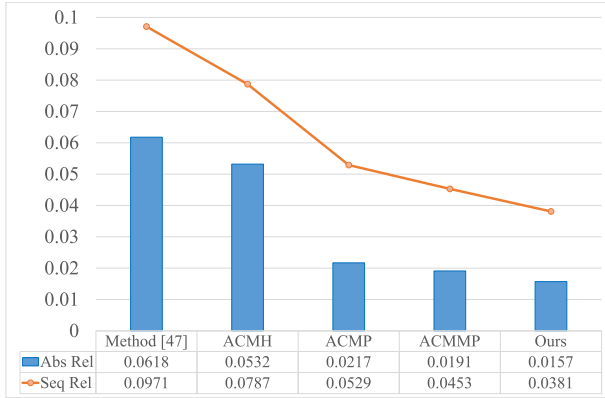


Fig. 12. Quantitative results from evaluation for the real images of a steep mountain scale model scene (lower is better).

exhibits the best performance among all compared methods. Considering the ground truth and the edges of the models in the depth maps, it is clear that the depth map obtained by the proposed method is more accurate than method [47], ACMH method, ACMP method, and ACMMP method.

2) *Model of a Steep Mountain*: The second scene contained a steep mountain with a peak height of 65 mm relative to the base. We imaged this scene from a vertical height of 800 mm and moved the camera at a baseline length of 5 mm; the B/H ratio was 0.00625.

The results presented in Fig. 10 indicate that the proposed method can improve the completeness of image depth estimations. The surface of the mountain is relatively smooth, which is particularly challenging to elucidate over poorly textured regions. Overall, the depth obtained by the developed method has fewer noisy regions on the low-textured areas and was more complete than reference methods.

3) *Quantitative Evaluation*: We estimated the quality of the depth map by calculating the absolute relative error (Abs Rel) and the square relative difference (Sq Rel), which are calculated as follows:

$$\begin{aligned} \text{AbsRel.} &= \frac{1}{n_y} \sum \frac{|y_{\text{pred}} - y_{\text{gt}}|}{|y_{\text{gt}}|} \\ \text{SeqRel.} &= \frac{1}{n_y} \sum \frac{(y_{\text{pred}} - y_{\text{gt}})^2}{y_{\text{gt}}^2}. \end{aligned} \quad (40)$$

In the formulas above, n_y is the number of pixels with the reference depth value, y_{pred} is the value of the predicted depth map, and y_{gt} is the depth value of the ground truth.

Figs. 11 and 12 show the depth estimation results from imaging the two scale model scenes. The method described herein outperforms state-of-the-art depth estimation methods for both datasets in terms of the two metrics, i.e., Abs Rel and Sq Rel. The accuracies of depth estimation obtained by our method, the ACMP method, and the ACMMP method exceeds that of method [47] and ACMH method, and our method estimates the detailed areas better than the ACMP method and ACMMP method. Outliers are often observed in the depth estimations performed by method [47], which leads to poor performance in the evaluated metrics because they are sensitive to outliers.

V. CONCLUSION

In this article, we have presented a precise and robust depth estimation method for UAV consecutive frames, which usually only satisfy narrow baseline stereo. To improve the depth estimation accuracy in real applications, 2-D image features and 3-D differential geometry characters are combined in image matching: By multiview geometry theory, it is proved that two corresponding local patches in two consecutive frames from UAV satisfy an affine transformation, no matter the height and pitch angle of the UAV. The affine transformation matrix and translation vector are calculated by the combination of normal curvatures, local Fourier moment constraints, spatial smoothness constraints, and the SLSF-PC method. The depth of the 3-D patch is calculated by the obtained affine transformation matrix and translation vector. Combining 2-D patch matching and 3-D patch differential geometry analysis makes the proposed method very robust, and SLSF-PC can well suppress illumination influence. Experimental results on the real UAV images and real images of scale model scenes indicate that the proposed method achieves state-of-the-art performance.

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